

SHETH MVLU COLLEGE

SUBJECT:- WSN

Aim:

Mote-to-PC Serial Communication Simulation

Implement serial communication between a sensor mote and PC to monitor real-time data transmission.

Code:

Mote.ned:

```
package pc;

simple Mote
{
    parameters:
        double sendInterval @unit(s);
    gates:
        output out;
}
```

PC.ned:

```
package pc;

simple PC
{
    gates:
        input in;
}
```

SerialNetwork.ned:

```
package pc;

network SerialNetwork
{
    submodules:
        mote: Mote
        {
            parameters:
                sendInterval = 1s;
        }

        pc: PC;

    connections:
        mote.out --> pc.in;
}
```

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Mote.cc:

```
#include <omnetpp.h>
```

```
class Mote : public cSimpleModule
```

```
{
```

```
    private:
```

```
        cMessage *timer;
```

```
    protected:
```

```
        virtual void initialize()
```

```
{
```

```
    timer = new cMessage("timer");
```

```
    scheduleAt(simTime()+1, timer);
```

```
}
```

```
        virtual void handleMessage(cMessage *msg)
```

```
{
```

```
    if(msg == timer)
```

```
{
```

```
        cMessage *data = new cMessage("Temperature=25C");
```

```
        EV << "Mote Sending Data\n";
```

```
        send(data, "out");
```

```
        scheduleAt(simTime()+1, timer);
```

```
    }
```

```
}
```

```
        virtual ~Mote()
```

```
{
```

```
    cancelAndDelete(timer);
```

```
}
```

```
};
```

```
Define_Module(Mote);
```

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PC.cc:

```
#include <omnetpp.h>

class PC : public cSimpleModule
{
    protected:

        virtual void handleMessage(cMessage *msg)
        {
            EV << "PC Received: "
                << msg->getName()
                << "\n";

            delete msg;
        }
};

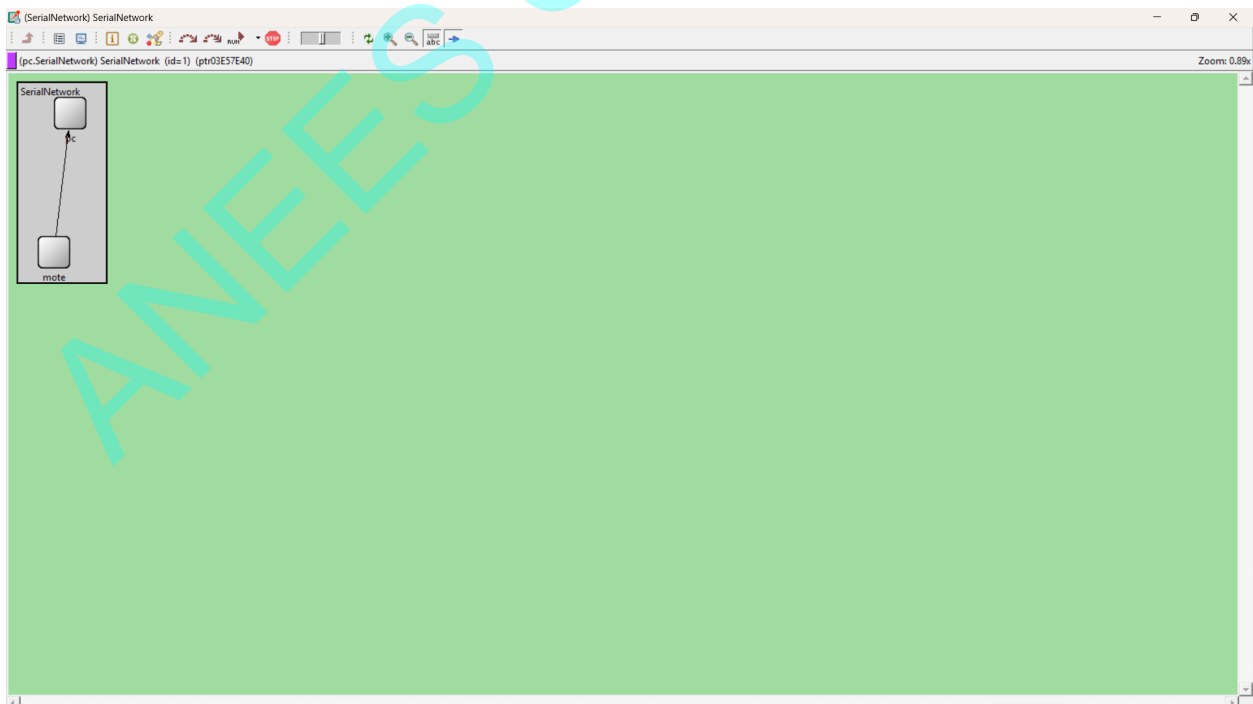
Define_Module(PC);
```

omnetpp.ini:

```
[General]
network = pc.SerialNetwork
sim-time-limit = 20s
```

Output:

Serial Network Topology:



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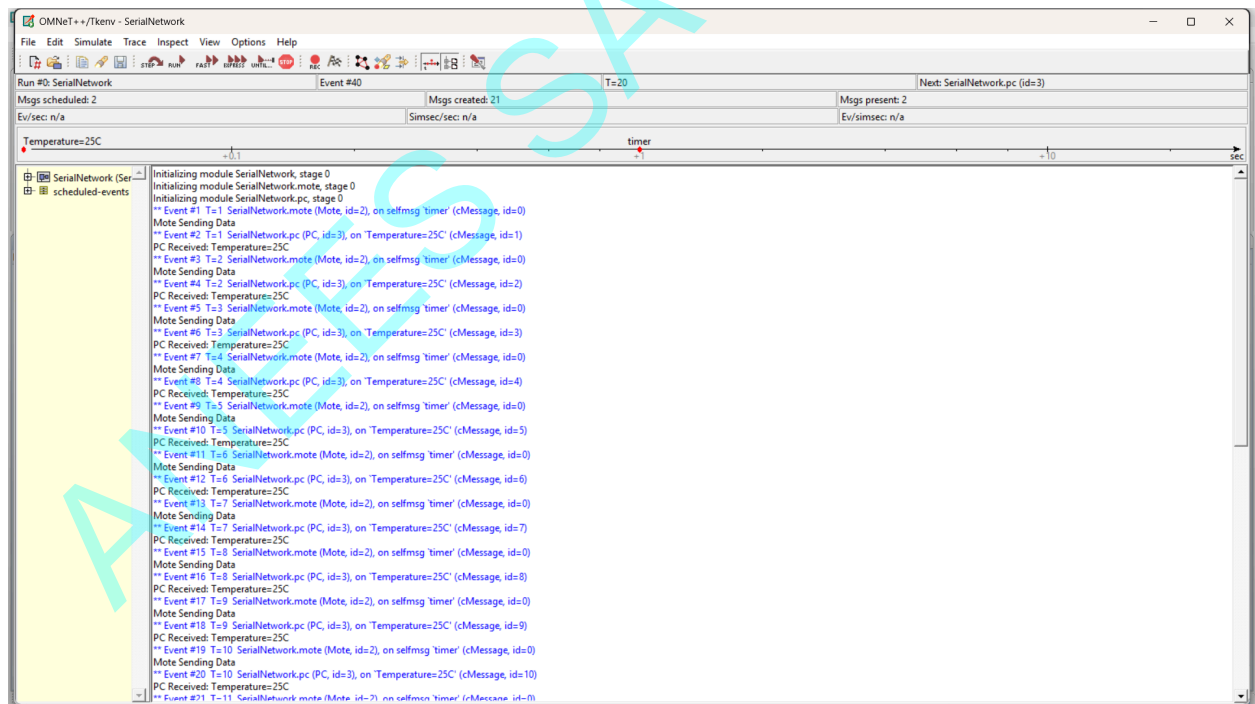
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Simulation Execution:



Mote Sending Data Log

PC Receiving Data Log



Result

The sensor mote successfully transmitted data to the PC through a serial communication link. The PC received and displayed the transmitted data in real time during the simulation.